

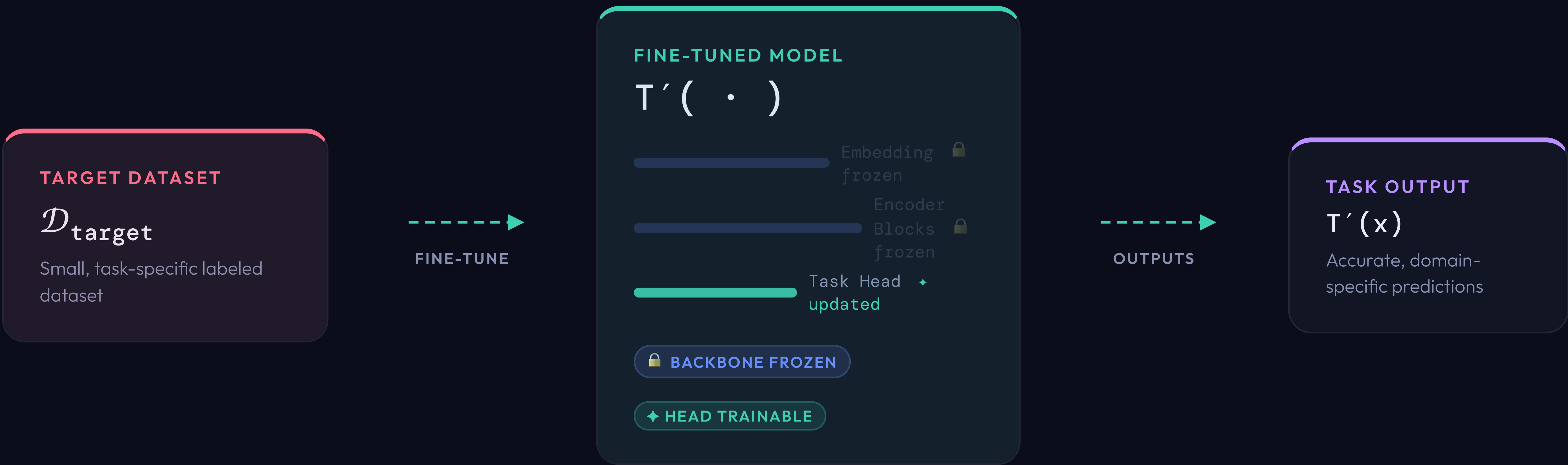
Transfer Learning & Fine-Tuning

KNOWLEDGE TRANSFER FROM A PRE-TRAINED MODEL $T(\cdot)$ TO A TASK-SPECIFIC MODEL $T'(\cdot)$

PHASE 1 — PRE-TRAINING ON LARGE CORPUS



PHASE 2 — FINE-TUNING ON TARGET TASK



- Large source dataset $\mathcal{D}_{\text{source}}$
- Pre-trained model $T(\cdot)$
- Small target dataset $\mathcal{D}_{\text{target}}$
- Fine-tuned model $T'(\cdot)$
- Frozen layers (transferred)
- Trainable (task-specific) head